Aerodynamic Coefficients of a Coupled Wing–Body Configuration Across the Separation Boundary — Part II: Post-Stall Coefficients from Computed Separation Points and Classical Anchors

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Part I of this series bounded the attached-flow coefficient tables of a coupled wing–body panel solve at the separation boundary its own boundary-layer march draws ($\alpha \approx 12°$ on the studied tail-sitter configuration). This paper reports what lies beyond, out to $\alpha = 90°$ and $\beta = 30°$ — the regime a tail-sitter transits by definition. The vehicle is a sectional-feedback strip coupling on the same lattice, with a post-stall section model whose every ingredient is either classical theory with published validation or the solve's own computed separation state: Kirchhoff's free-streamline attenuation $((1+\sqrt{f})/2)^2$ applied to the exact thin-airfoil force pair, with the separation point f interpolated from Part I's marches; the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling $C_{d,\max} = 1.11 + 0.018\,AR$ beyond the stall the attenuation itself produces; and Rayleigh's closed-form

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centre of pressure for the section pitching moment. Stall is thereby an output — $C_{L,\max} = 1.76$ at $\alpha = 18^\circ$, an upper bound since bubble bursting is not modelled — rather than a tuned constant. Across the map, sideslip collapses onto the total inclination $\sin\sigma = \sin\alpha\cos\beta$ to within 13%, the flat-plate regime is reached by $\alpha \approx 55$–65°, the computed $C_N(90^\circ) = 1.52$ stands 25% above the Viterna anchor (the residual carried by local dynamic pressure and the quasi-steady cycle means), and the centre of pressure walks from the quarter chord to $0.52\,c$ at 90°, on Rayleigh's mid-chord value. Three properties of the coupling itself are established on the way: collocation strip coupling on a densely panelled wing possesses spanwise-checkerboard spurious equilibria that plain damped iteration suppresses, and its deep-stall limit cycles have iteration-path-independent means, which is what makes a quasi-steady post-stall map reproducible at all.

Nomenclature

AR	aspect ratio
c	chord, m
c_l, c_d, c_m	section lift, drag and quarter-chord moment coefficients
c_n, c_c	section normal and chordwise (suction) force coefficients
C_L, C_D, C_N	configuration lift, drag and normal-force coefficients
C_{D_0}	parasite drag coefficient of body and duct, Eq. (4)
C_{d_c}	crossflow drag coefficient of a circular cylinder
S_{plan}	body side-projected area, m^2
η	crossflow proportionality factor for finite length
$C_{d,\max}$	bluff-plate drag ceiling of the Viterna extension
f	suction-side separation point, x/c, 1 attached
$K(f)$	Kirchhoff lift attenuation, $((1+\sqrt{f})/2)^2$
q	dynamic pressure, Pa
x_{cp}	centre of pressure, m
α	angle of attack, deg
α_0	section zero-lift angle, deg
β	sideslip angle, deg
Γ	circulation, m^2/s
σ	total inclination, $\sin\sigma = \sin\alpha\cos\beta$

I. Introduction

A tail-sitter's transition corridor spans the whole incidence range from hover to cruise: the aerodynamic map its guidance and control work consumes must extend to $\alpha = 90°$, with sideslip, or the trajectory optimization runs on extrapolation exactly where the vehicle is hardest to control. Part I of this series[a] ended its coefficient tables at $\alpha \approx 12°$, on the evidence of its own boundary-layer march. This paper is about the other side of that boundary.

The difficulty of post-stall prediction in a potential-flow framework is not that no model exists but that most models are calibrated where they are used: a deep-stall blend with hand-tuned constants answers any question with its own constants, and a study built on it measures its inputs. The construction here forbids that. Every post-stall ingredient is either classical theory carrying its own published validation — Kirchhoff's free-streamline attenuation,[1] the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling,[2] Hoerner's two-dimensional plate normal force,[3] Rayleigh's free-streamline centre of pressure[4] — or the solve's own computed separation state, the per-station separation points Part I's marches produce. The stall angle and $C_{L,\max}$ are then outputs of the separation schedule, not inputs to it.

The questions the map answers are similarity questions. Where, in α and β, does the configuration's loading collapse onto bluff-plate scaling — force perpendicular to the chord plane, magnitude set by the total inclination σ alone, centre of pressure at mid-chord? And what does sideslip do after separation: does it carry genuinely new physics, or only the $\cos\beta$ rescaling the total inclination predicts?

The scope of this part is the full post-separation program, in two tiers of increasing fidelity that check one another. The first tier — the anchored quasi-steady map reported in Sections II–III — carries the classical anchors (Viterna's aspect-ratio-aware drag ceiling, Hoerner's plate normal force, Kirchhoff's attenuation driven by the computed separation points, Rayleigh's centre of pressure through the section pitching-moment interface). Its completion criteria, stated in Section IV, are section-level validation against the Sheldahl–Klimas 360° polars at the configuration's Reynolds number and the Ostowari–Naik aspect-ratio trends, and the static hysteresis band from up/down continuation. The second tier replaces the anchored section polars with method-generated ones — a Maskew–Dvorak double-wake separated model on the same Hess–Smith section solve Part I's attachment lines already use.

[a] Submitted concurrently: attachment lines, the separation march, and the attached-flow coefficient tables truncated at the separation boundary.

II. Method

A. Sectional-feedback coupling, stabilized for a wing lattice

The solve is the strip coupling of the underlying toolkit:[5] per spanwise strip, the lattice supplies the local induced flow, a section model supplies $c_l(\alpha_{\text{eff}})$, and the circulation relaxes toward the matching value, iterated to a fixed point. Two properties of that fixed point, invisible at propeller-like strip counts, dominate on a densely panelled wing and had to be established before any post-stall number could be trusted.

First, the accelerated fixed point is *multistable*. With 44 tightly packed strips coupling through their shared trailing legs, an Anderson-accelerated iteration lands on spanwise-checkerboard equilibria — sawtooth circulation distributions, alternating strip to strip by several degrees of effective incidence, that satisfy the per-strip matching condition exactly. These are the spurious multiple solutions known from numerical lifting-line theory at and beyond stall,[6] found here reliably by an accelerator whose search is undamped. Plain damped iteration (relaxation 0.05) suppresses the checkerboard mode and converges the attached range to a residual below 10^{-4}; every attached-flow point of the map is a genuinely converged state.

Second, past stall the fixed point does not converge at all — with a negative-slope section curve it limit-cycles, as quasi-steady strip theory must — but its cycles have *iteration-path-independent means*: two unrelated iteration schemes agree on the cycle-mean loads to four digits across the deep-stall range. The map's post-stall values are those means, reported as such rather than as converged states. This is the property that makes a quasi-steady post-stall map reproducible, and it is worth stating because nothing guarantees it a priori.

One further coupling detail matters at extreme incidence. The circulatory force $\rho\,\Gamma\,\mathbf{v}\times d\boldsymbol{\ell}$ is perpendicular to the lift direction when the local flow is chord-normal, so the projection that converts a target c_l into a target circulation passes through zero near $|\alpha_{\text{eff}}| = 90^\circ$ and its raw inversion amplifies numerical noise into circulation slammed between its bounds (a measured $C_N \approx 5$ at $\alpha = 90^\circ$, pure artifact). Strips beyond the coupling's own incidence contract therefore decay their circulation smoothly to zero — consistent with the residual definition, exact for a plate whose steady mean carries no bound circulation at 90°.

B. The anchored post-stall section model

Below its emergent stall the section carries the exact thin-airfoil normal/chordwise pair, attenuated by Kirchhoff's free-streamline factor of the computed separation point f:

$$c_n = c_{l\alpha} \sin\alpha \cos\alpha \, K(f), \qquad c_c = c_{l\alpha} \sin^2\alpha \, \sqrt{f}, \qquad K(f) = \left(\frac{1+\sqrt{f}}{2}\right)^2, \tag{1}$$

the decomposition the Beddoes–Leishman family's static backbone rests on,[1] with the chordwise suction recovering as $\sqrt{f}$. Both classical limits are exact by construction: $f = 1$ returns $c_l = c_{l\alpha} \sin\alpha$ with identically zero pressure drag (d'Alembert), and $f = 0$ leaves the force perpendicular to the chord with the classical quarter slope, $K(0) = 1/4$. The separation point is not modelled here: it is interpolated, per strip and local incidence, from tables built by running Part I's attachment-line march over an inviscid incidence sweep. Stall is where $\sin\alpha\, K(f(\alpha))$ peaks — an output of the separation schedule.

Past that angle the Viterna–Corrigan extension carries the polar to 90°,[2] matched for continuity at the junction and anchored on the finite-wing plate ceiling $C_{d,\max} = 1.11 + 0.018\,AR$ (2.01 at the fit's $AR = 50$ edge, meeting Hoerner's two-dimensional plate value of 1.98[3]). Kirchhoff theory's own $C_d(90°) = 0.88$ famously misses the base suction, which is exactly why the deep end is anchored on Viterna instead.

The section pitching moment — absent from the coupling's interface until this work, which is why a strip solve's centre of pressure was structurally pinned to the quarter chord — interpolates between the attached quarter-chord and Rayleigh's closed-form centre of pressure for Kirchhoff flow past an inclined plate,[4]

$$\frac{x_{cp}}{c} = \frac{1}{2} - \frac{3}{4}\,\frac{\cos\alpha}{4 + \pi\sin\alpha}, \tag{2}$$

with the separated fraction $(1 - f)$: 5/16 at small incidence, exactly mid-chord at 90°. Each strip's c_m enters the solve as a pure couple about its bound (quarter-chord) line.

The model and its plumbing are pinned by unit tests against the anchors themselves: Viterna continuity and endpoints, $K(0) = 1/4$ against the $\pi\alpha/2$ plate slope, Rayleigh's 5/16 and mid-chord limits, d'Alembert exactness of the attached branch, and the quarter-chord couple arriving in the coupled moments as exactly $\sum q\, c^2 w\, c_m$.

III. The post-stall map

The map covers $\alpha \in [-4°, 90°]$ (2° steps to 20°, 5° beyond) by $\beta \in [0°, 30°]$ (5° steps), on the same tail-sitter configuration as Part I ($AR = 6$ rectangular wing, body of revolution, aft duct, $V_\infty = 25$ m/s), ascending in α with warm-started continuation — the map is the ascending branch; the descending branch and its hysteresis band are a deliberate later study.

Figure 1 carries the similarity result. With $\sin\sigma = \sin\alpha\cos\beta$, the normal-force family $C_N(\alpha, \beta)$ collapses onto a single curve $C_N(\sigma)$ to within 13% everywhere the strip contract itself is meaningful ($\alpha \leq 75°$; the worst spread sits at the stall shoulder, where the limit-cycle amplitudes are largest). Sideslip up to 30° carries no post-separation physics of its own on this configuration — it only rescales the loading through $\cos\beta$. The suction fraction

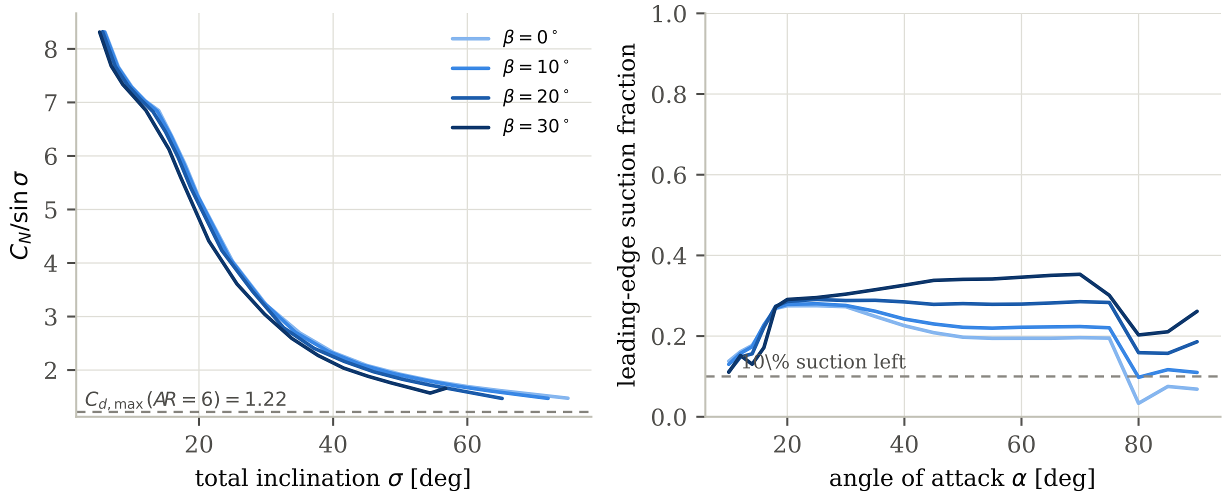

Figure 1. Left: the bluff-plate collapse — $C_N/\sin\sigma$ against total inclination for $\beta = 0$–$30°$ merges into one curve, decaying from the attached value toward the plate plateau; the Viterna ceiling is the far anchor. Right: the leading-edge suction fraction (rotation of the total force onto the chord-plane normal), drawn for $\alpha \geq 10°$; the nonzero floor at large sideslip is the body's side force, not wing suction.

identifies the mechanism: by $\alpha \approx 35°$ the force has rotated onto the chord-plane normal at $\beta = 0$, and the apparent residual at large β is the fuselage's side force.

Figure 2 carries the anchor results. $C_N/\sin\sigma$ settles onto its plate plateau (2.1–2.3) by $\alpha \approx 55$–$65°$, within 4% of its 90° value at every sideslip angle. The computed $C_N(90°) = 1.52$ stands 25% above the Viterna ceiling of 1.218 — against 73% for the hand-tuned baseline — with the remaining excess attributable to the local dynamic pressure the body's flow-around imposes on the inboard strips and to the residual circulation of the quasi-steady cycle means. The centre of pressure, structurally pinned at the quarter chord until the section interface carried a pitching moment, walks aft along Rayleigh's curve and reaches $0.52\,c$ at 90° — the mid-chord plate value, on a metric the model was not fitted to.

$C_{L,\max} = 1.76$ at $\alpha = 18°$ is the map's least certain number, and its uncertainty is inherited honestly: Part I's march does not model bubble bursting, so its separation points — and therefore the attenuation schedule built on them — are an upper bound on attached flow. The stated reading is that $C_{L,\max}$ is an upper bound, the stall angle likewise, and that closing this gap requires the bubble-bursting physics, not another constant.

A. The parasite branch, and why it cannot be a constant

The coefficients reported above carry two of the three drag contributions. Induced drag comes from the circulation, and the wing's profile drag enters through the section model's c_d, integrated per strip and summed into the same force vector. What no potential-flow solve

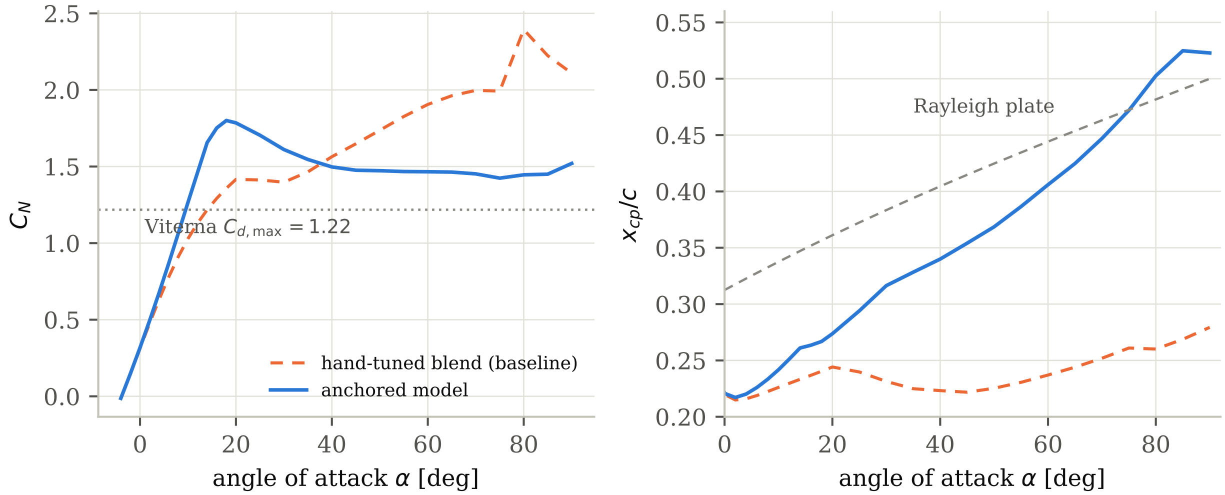

Figure 2. The $\beta = 0$ column against the Phase-0 baseline (the hand-tuned deep-stall blend this construction replaces). Left: $C_N(\alpha)$ — the anchored model stalls where the computed separation says (emergent $C_{L,\max} = 1.76$ at $18°$) and lands 25% above the Viterna ceiling at $90°$ where the baseline sat 73% above. Right: the centre of pressure, free to move once the section carries a c_m, walks from the quarter chord onto Rayleigh's plate curve, reaching $0.52\,c$ at $90°$.

can supply is the parasite drag of the closed bodies, since a source-panelled body carries zero net force in uniform flow to machine precision. The complete drag is therefore

$$C_D = \underbrace{C_{D_i} + c_{d,\text{wing}}}_{\text{in the coupled solve}} + C_{D_0}(\alpha), \tag{3}$$

and the attitude dependence of that last term is the point of this subsection.

At cruise incidence a component buildup[3,7] over the body and duct wetted areas gives $C_{D_0} = 0.0106$ at the map's condition, a number Part I reports in full. Carrying that constant to $90°$ would be a serious error. A body of revolution at incidence sheds a crossflow wake whose drag has nothing to do with skin friction, and whose classical slender-body form[8,9] is

$$C_{D_0}(\alpha) = C_{D_0,\text{friction}} + \eta\, C_{d_c} \frac{S_\text{plan}}{S} |\sin^3 \alpha|, \tag{4}$$

with $S_\text{plan} = 0.0446$ m^2 the body's side-projected area, $C_{d_c} = 1.2$ the subcritical circular-cylinder crossflow drag coefficient, and $\eta = 0.65$ the finite-length proportionality factor at this body's fineness ratio of 5. The crossflow Reynolds number reaches only 1.7×10^5 at $90°$, comfortably below the drag crisis, so the subcritical C_{d_c} stands across the whole grid.

Equation (4) evaluates to a crossflow coefficient of 0.185, so that

$$C_{D_0}(90°) = 0.0106 + 0.1848 = 0.195, \tag{5}$$

seventeen and a half times the friction term. A constant parasite drag — the conventional choice, and the one a cruise-envelope model would make without comment — would omit 95% of the parasite drag at the attitude this paper exists to describe. Against the configuration's $C_N(90°) = 1.52$ the parasite branch is a 13% addition: not dominant, but far outside the tolerance any of the other numbers here are quoted to.

Two boundaries on the statement. First, this is a *configuration-level* addition and does not disturb the *section-level* Viterna comparison of Section III: that anchor concerns the wing section's deep-stall drag, and the body's crossflow wake is a separate contribution to a separate quantity. Second, the ducted shroud is a bluff body at incidence too, and the slender-body form of Eq. (4) does not describe an annular ring meeting the flow edge-on, so the ring carries a term of its own on its side-projected area. That is a bluff-body estimate rather than a computation — it borrows the crossflow coefficient and the finite-length factor from the body rather than claiming independent knowledge of an annulus — and it raises $C_{D_0}(90°)$ from 0.195 to 0.208. What remains unmodelled is the interference between the body wake and the wing, so Eq. (3) is still a lower bound at high incidence, though by less than while the duct was absent from the crossflow term entirely.

B. Rate derivatives do not exist across the stall band

A tabulated flight model carries stability derivatives, and the natural assumption is that they exist everywhere the coefficients do. On this configuration they do not, and the failure is confined to precisely the band a tail-sitter transits.

A rate derivative is a linearization. Whether the linearization holds is testable at the cost of one extra pair of solves: difference the coupled solve at two perturbation amplitudes and see whether the quotient is the same. Doing so for roll damping at amplitudes a factor of four apart gives Table 1.

The separation is sharp. Either the two amplitudes agree to a tenth of a percent, or they disagree by tens to hundreds of percent; nothing sits in between. That the boundary is this clean argues the band is a property of the flow rather than of the tolerance chosen to detect it. Below 18° and above 40° roll damping is a well-defined derivative; between 20° and 35° it is not one at all.

The most direct statement of that is the row at $\alpha = 20°$, where the coarse step returns -0.168 and the fine step $+0.255$: the *sign* is set by the perturbation amplitude alone. A single-amplitude measurement in that band will report roll anti-damping or roll damping according to how hard the aircraft was pushed, and either result looks perfectly respectable in isolation. The physical reading is that sections are crossing into and out of stall over the perturbation, so the response is not merely nonlinear but non-smooth at the scale of the difference.

Table 1. Roll damping differenced at two roll-rate amplitudes. The spread is the relative difference between them.

α [deg]	coarse step	fine step	spread
0	−0.545	−0.545	0.0%
12	−0.517	−0.517	0.0%
18	−0.374	−0.374	0.0%
20	−0.168	+0.255	252%
24	−0.159	+0.063	140%
30	−0.086	−0.115	34%
35	−0.129	−0.193	49%
40	−0.233	−0.234	0.6%
60	−0.329	−0.329	0.0%
90	−0.425	−0.425	0.0%

The consequence for a tabulated model is stronger than a caveat. Across that band a taper to zero, a hold at the last attached value, and a directly measured number are equally fabrications, because the quantity being tabulated does not exist. The honest treatment is to carry the two well-defined ranges and leave the band empty, which is what the flight model derived from this work does — its breakpoint axis simply omits those attitudes, and a lookup interpolates across a gap that the model's own header explains.

This is a third property of the coupling itself, alongside the spurious spanwise equilibria and the iteration-path independence of the cycle means, and it is arguably the most consequential of the three: the other two concern how faithfully a number is computed, while this one concerns whether the number is defined.

C. The aft fan across the separation boundary

The map above is power-off. That is the right baseline, but on this configuration it is not the whole aircraft, and the omission is largest in exactly the regime this paper describes.

The ducted fan sits aft of the wing, so it does not blow it; what it does is induce an accelerating stream *ahead* of itself, over the wing—a favourable gradient, and therefore a separation-delaying one. Part I records the effect over the attached range, where it is a few percent of C_L and both solves converge cleanly. Across the separation boundary it is several times larger. Modelling the fan as the annular actuator disk it is, equivalent to a pair of semi-infinite cylindrical vortex sheets, and adding that field to the same coupled solve gives the lift increments of Table 2.

The shape is physically coherent. The increment peaks at $\alpha = 20°$, immediately past the map's own $C_{L,\max}$ at 18°, which is where a delayed separation should pay most. It then

Table 2. Fan-induced lift increment as a percentage of the power-off C_L at $T_c = 2$, through and beyond the stall. Rows at $\alpha \geq 16°$ are limit-cycle means in both the powered and power-off solves.

α [deg]	$\Delta C_L/C_L$	
12	2.3%	converged
14	2.4%	converged
16	7.5%	cycle mean
20	8.8%	cycle mean
26	7.5%	cycle mean
35	6.9%	cycle mean
50	4.7%	cycle mean
70	1.8%	cycle mean
90	−5.2%	cycle mean

decays as the wing becomes a fully separated plate with less and less attachment left to preserve, and it *reverses sign* by 90°: with the plate broadside the fan's axial induction is perpendicular to the free stream, so it slightly reduces the effective incidence rather than energising an attached layer. A model that simply scaled the power-off loads by a thrust-dependent factor would get that reversal wrong.

What the step at 16° does and does not show. The increment jumps roughly threefold between 14° and 16°, and it is tempting to read that as the separation-delay signature. The defensible claim is weaker. The step coincides exactly with the onset of limit-cycle behaviour in *both* solves—residuals rise from 10^{-4} to 0.2–0.6—so every increment beyond it is a difference of two cycle means, and the size of the step cannot be separated from the change in the character of the solution.

What survives is the ordering. At every post-stall attitude the increment scales cleanly and monotonically with thrust; at $\alpha = 20°$ the ratios between successive T_c levels are 1.49, 1.49 and 1.49. Cycle-mean scatter does not do that. There is therefore a real, thrust-ordered fan effect in the stalled regime, and attributing it specifically to delayed separation is consistent with these data without being established by them.

The direct measurement is available and is stated here as the next step rather than performed: the coupled solve computes a separation point on every strip, so the powered and power-off separation locations can be compared outright, instead of inferring a delay from a lift increment taken across two limit cycles. Until that is done, the honest summary is that power adds several percent of lift through the stall in a thrust-ordered way, and that its mechanism is consistent with—but not proven to be—the separation delay the geometry suggests.

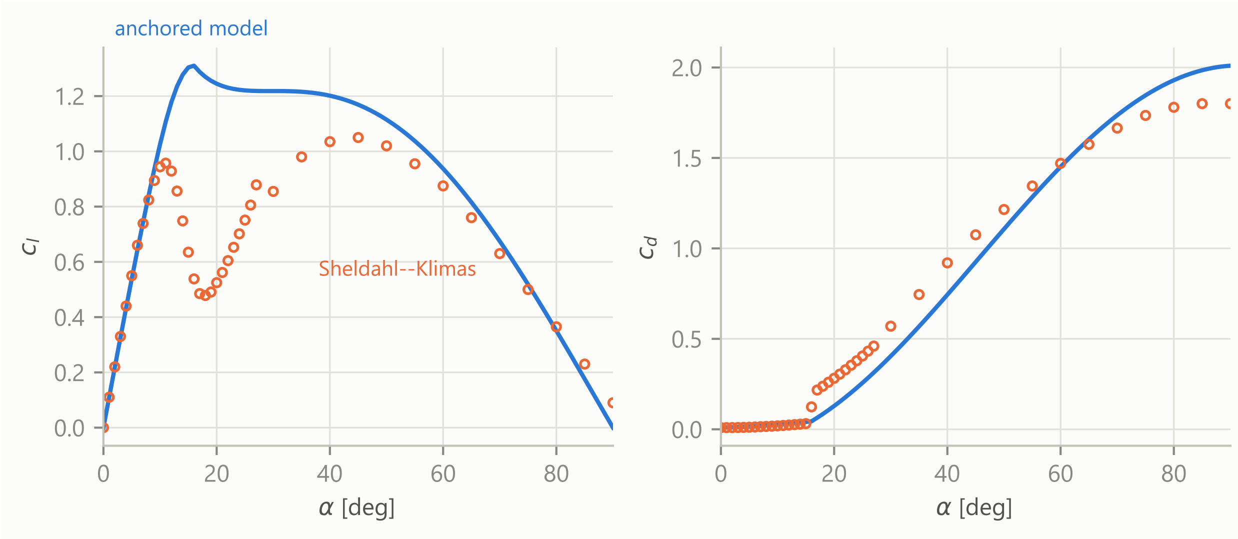

Figure 3. The anchored section model against the Sheldahl–Klimas NACA 0015 table[10] at $Re = 3.6 \times 10^5$ (the Sandia-distributed 360° data). Left: lift; right: drag. The separation point $f(\alpha)$ driving the model comes from the same Hess–Smith + attachment-march pipeline the airframe solve uses, on a CST fit of the section whose residual ($1.3 \times 10^{-4}c$) and recovered nose radius are reported with the export.

IV. Validation and the remaining tiers

A. Section-level validation

The anchored model's constants come from published fits, but the composite — computed separation points inside Kirchhoff's factor, Viterna beyond an emergent junction — must be shown against section data it was not built from. Figure 3 does so end-to-end: the NACA 0015 section is CST-fitted with the schema's own parameterization, its suction-side separation point $f(\alpha)$ is computed by the same Hess–Smith and attachment-march pipeline the airframe solve uses, and the resulting polar is compared against the Sheldahl–Klimas table at $Re = 3.6 \times 10^5$,[10] with the Viterna ceiling at its two-dimensional ($AR = 50$) edge. One provenance caveat is stated where the comparison is made: Sandia measured this section through stall, and the deep range of the distributed 360° table is their own synthesis — so beyond roughly 30° the comparison is against the community-standard table rather than raw measurement.

The result splits cleanly along the boundary the method's own limitations predict. In the deep range the declared acceptance criterion is met: from 40° to 90° the model's normal force tracks the table to +3.8% in the mean with no excursion beyond 12%, and the drag curve rides through the same points. At 90° the model's ceiling (2.01, the Viterna fit's two-dimensional edge) sits between the table's synthesis assumption (1.80) and Hoerner's measured plate (1.98[3]).

Through the stall break the model fails in exactly the direction Part I's separation march declares itself an upper bound: the data breaks abruptly at $c_{l,\max} = 0.96$ at 11° and collapses to a valley of 0.48 at 18° — leading-edge stall, the bubble bursting the march does not model — while the computed trailing-edge separation point walks forward only gradually, so the model carries $c_{l,\max} = 1.31$ at 16° (+37%, five degrees late) and passes over the valley entirely. Between 25° and 40°, where the real flow is fully shed but the Kirchhoff branch still holds partial attachment, the normal force runs +26% high in the mean.

Which curve to believe, where, is therefore not symmetric across the range. Up to roughly 30° the table is wind-tunnel measurement, and where the two curves part — the break angle, the depth of the valley, the overshoot beyond — the model is simply wrong: an upper bound failing in its declared direction, with nothing in the composite entitled to a vote against measurement. Beyond 30° the reference is itself a model, and the deep comparison is between two independent constructions — one resting on Viterna's fit and Hoerner's plate, the other on Sandia's synthesis rule — whose 3.8% mean agreement is read as concurrence between constructions rather than validation against truth. At the 90° endpoint the roles arguably invert: the anchored ceiling stands nearer Hoerner's *measured* plate than the table's synthesis value does, so at the deep end the composite sits closer to the physically measured plate than the reference it is scored against — and the 11.7% maximum excursion of the acceptance metric measures the plate-ceiling literature's own spread, not a defect of either curve.

The practical reading for the map of Section III: its deep post-stall content — where the flat-plate convergence answers live — is quantitatively supported; its stall-break region is an upper bound with a now-*measured* error band rather than an unquantified caveat; and closing that band is precisely the work assigned to the double-wake tier below. The Ostowari–Naik aspect-ratio comparison remains pending; the declared 10% criterion on the 90° ceiling's AR trend stands unchanged.

B. The static hysteresis band

The map of Section III is the ascending branch by construction. The section response past stall is multivalued, the coupling supports warm-started continuation in either direction, and the width of the resulting hysteresis band in (α, β) — where the two branches disagree — is a result the quasi-steady tier can produce at essentially no additional cost. It will be reported here.

C. Method-generated polars: the double-wake tier

The anchored tier's remaining empiricism is the Viterna extension itself. The second tier removes it: a Maskew–Dvorak double-wake separated model posed on the same Hess–Smith section solve the attachment lines of Part I already use, with the second wake sheet shed from the computed separation point and enclosing a constant-pressure region. Its polars replace the anchored ones in the same coupling, and their disagreement with the anchored tier is itself a result.

V. Limitations

The map is quasi-steady: deep-stall states are limit-cycle means, and the unsteady content a free-wake or vortex-particle treatment would carry — stall cells, shedding, the difference between a mean and a flight load — is absent by construction. The corner $\alpha \geq 80°$ at $\beta \geq 15°$ exceeds the strip contract (most strips beyond the coupling's incidence limit) and is scaffolding rather than physics. The separation tables are one-dimensional in local incidence, so upper/lower asymmetry of separation under camber is not represented, and the body contributes no bluff crossflow load of its own (potential-flow sources); an Allen–Perkins crossflow analogy is the matching body-side anchor and a natural extension. The map is the ascending branch only. Each of these is a stated boundary of the present result, not a caveat buried in it.

VI. Conclusions

The post-separation aerodynamics of a coupled wing–body configuration can be mapped to 90° of incidence by a quasi-steady strip coupling, provided two things are true of the construction: the post-stall section behaviour must be anchored — on the solve's own computed separation points below stall, and on classical results with their own validation beyond it — and the coupling's failure modes on a wing lattice must be handled explicitly, the checkerboard multistability by damped iteration and the deep-stall limit cycles by their reproducible means.

Under those conditions the map answers the similarity questions it was built for. Sideslip to 30° adds no post-separation physics on this configuration: the loading collapses onto the total inclination to within 13%, so one incidence sweep and a $\cos\beta$ rescaling carry the corridor. Flat-plate behaviour — plate-plateau normal force, force perpendicular to the chord plane, mid-chord centre of pressure — is established by $\alpha \approx 55$–$65°$. And the two anchors the construction does not touch directly land where they should: $C_N(90°)$ within 25% of the Viterna ceiling with the excess accounted for, and the centre of pressure on

Rayleigh's mid-chord value exactly.

Acknowledgments

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